Week 6 · Day 04 homework

Grow the evaluation set to 100 cases, and write the labelling guidelines

Forward Deployed Engineer Program · Phase 2 · Mithran Sadasivam · 2026-11-06

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All numbers come from saved runs in engagement/eval/runs/ on the real model. The 30 new synthetic cases were generated with python -m scripts.make_synthetic and checked against the v2 guidelines.

1 · Labelling guidelines v2

For anyone labelling UC1 proof-of-address extractions. Written so a second analyst who was not in the Day 4 session can label and agree with Tom.

v2 keeps everything from v1 (names and dates, as agreed with Tom in the two labelling rounds) and adds address and issuer, one real example per field, the hard cases and what to do when unsure.

The rule

A value is **correct** when it is exactly what the document prints for that field, nothing missing and nothing added, and it is the right item on the page (the issue date, not the payment date; every person addressed, not just the first). Judge the value against the page, never against the application and never against what the rules would do with it.

Field by field

Field	Correct	Wrong	Example from the evaluation set
Full name	Everyone the document is addressed to, titles included, joined as printed (`Mr A Shah & Mrs P Shah`). Initials exactly as printed.	A joint bill with a name missing. A name with the title dropped or an initial expanded.	**L03 · D04-033** `Mr Arun Shah`: **wrong**, the bill is addressed to Mr Arun Shah & Mrs Priti Shah. **L02 · D03-008** `Mr J. A. Kowalski`: **correct**, initials as printed.
Address	The postal lines in the printed order, flat number and postcode included, joined with ", ". If the page prints `number street, Flat n`, that is the correct order even when the application says `Flat n, number street`.	A line, flat number or postcode missing, or the lines re-ordered to match the application.	**L11 · D04-041** `8 Brunswick Road, Flat 5, Gloucester, GL1 1JJ`: **correct** as printed (the application writes it `Flat 5, 8 Brunswick Road`). *No wrong address has appeared in the 100 cases so far; the wrong column is the rule, not an observed example.*
Statement date	The date beside an issue label: "Statement date", "Date of issue", "Bill date", "Issued". Written as printed.	A payment due date ("Payment due", "Due by", "Pay by"), a billing period or a transaction date, even when it is printed first.	**L01 · D04-031** `03 Oct 2026`: **wrong**, it is the payment due date. **L08 · D04-037** `01 Oct 2026`: **correct**, "Date of issue" on a council tax bill.
Issuer	The organisation's name as printed, legal suffix included (`LUMENVALE ENERGY LTD`, `Northmere Bank plc`), capitals as printed. The name only: stop before the service tagline (`Water & Sewerage`, `Electricity & Gas`).	Another organisation, a name that is not printed, or the name with the tagline run on.	**L10 · D04-036** `LUMENVALE ENERGY LTD`: **correct** as printed. **D04-043** the model's `Harrowgate Water & Sewerage` against the label `Harrowgate Water`: **wrong**, the tagline is not part of the name.

What a label is not

- **Not a decision.** D04-036's issuer is correct as printed even though the approved-issuer rule rejects `LUMENVALE ENERGY LTD`. That is a rules problem, not a label problem. Never bend a label to make a rule pass.
- **Not a measure of confidence.** Correct or wrong; "probably" is not a label (see "When you are not sure").
- **Not a view of the application.** Do not open the application form to decide a label. Whether the name or address matches the applicant is a separate check.
- **Not a check on the scan.** Label what the document says. If the OCR misread a digit (D03-009 prints 18 Sep, OCR read 16 Sep) the label stays the true value, and the miss is a failure of reading, which is a different fix.

Hard cases, decided

Case	The question	The decision, and who made it
D04-037	Is "Date of issue" on a council tax bill the statement date?	Yes, statement date. Tom (label `01 Oct 2026` correct).
D04-031, D04-034, D04-038	Two dates are printed and the payment date comes first: which is the statement date?	The date beside the issue label. The payment date is wrong. Tom. In round 1 (guidelines v0) the saved labels called D04-031 and D04-034 correct and Tom called them wrong; round 2 (v1) fixed both.

Case	The question	The decision, and who made it
D04-033, D04-040, D04-046, D04-048	A joint bill: is the first name alone enough?	No. Both names must be recorded. Tom ("Joint bill: both names must be recorded"). `Mr Tariq Aziz` (D04-040) was still a disagreement with Tom in round 2 (`v1`), so the rule is stated in full here.
D03-008	Initials: expand or keep as printed?	As printed (`Mr J. A. Kowalski`). Tom.
D04-036, D04-049	Issuer printed with a legal suffix or in capitals: same company, which form is the label?	Same company; the label is the name as printed, suffix included. Tom ("Same company: its legal name").
D04-041, D04-045	Address printed `number street, Flat n`: is that the same flat as `Flat n, number street`?	Same flat; the label keeps the printed order. Tom ("Same flat, written the other way round").
D03-010, D04-043	Issuer header reads `Harrowgate Water Water & Sewerage`: is the tagline part of the issuer?	No: `Harrowgate Water`. Tom's label; the extraction run copied the tagline in 4 of 50 real cases with extract_v2 and 6 with extract. Written down by me after the Day 4 homework runs, so Tom should confirm.
S-143	A synthetic case is both a joint bill and an issuer variant (`Fenwick Gas & Power plc`, two names). Which slice?	It is labelled issuer_variant but the name label records both people (joint rule applies). Me, from the guidelines; Tom to confirm.
S-140, S-143	A synthetic statement date after the application date (05 Nov 2026, 01 Dec 2026). Does that change the label?	No. The label is what the page prints; whether the date passes the three-month rule is a rules question. Me.

When you are not sure

Do not guess and do not pick the likelier value. Mark the item **unsure**, write one line saying what you could not decide, and leave it out of the scores until it is answered. Ask Tom, in the labelling thread, with the case id and a screenshot or the OCR words. Tom's answer becomes a new row in the hard-cases table above, with his name, so the next labeller does not have to ask again. If a synthetic label makes you hesitate, the guidelines are missing a line: add it, then fix or delete the case and record which in `evals/synthetic/LABEL_NOTES.md`.

2 · The 100 cases and the label spot-check

50 cases, all written by the model through `python -m scripts.make_synthetic` (real model, not the scripted one), text and OCR words only.

Slice	Command	Cases
two_dates	`make_synthetic two_dates --n 20`	S-101 to S-120
joint_names	`make_synthetic joint_names --n 15`	S-121 to S-135
issuer_variant	`make_synthetic issuer_variant --n 8`	S-136 to S-143
address_order	`make_synthetic address_order --n 7`	S-144 to S-150

Spot-check against docs/eval/LABELLING_GUIDELINES.md (v2)

I checked 8 cases as text against the guidelines, then checked all 30 new labels with a script (every name, date and issuer appears in the printed words; every address appears in printed order).

Case	Slice	Check	Result
S-121	joint_names	both names in the label	matches
S-125	joint_names	both names in the label	matches
S-133	joint_names	both names in the label	matches
S-136	issuer_variant	`Lumenvale Energy Ltd` suffix kept	matches
S-137	issuer_variant	`HARROWGATE WATER` capitals as printed	matches
S-140	issuer_variant	`Northmere Bank plc` as printed	matches
S-144	address_order	printed `23 Acacia Avenue, Flat 4`, label keeps that order	matches
S-147	address_order	printed `12 Pine Lane, Flat 3`, label keeps that order	matches

Labels fixed or cases deleted

No label was edited. One event to record:

- The first generation of S-136 to S-150 (issuer_variant and address_order) disappeared from `evals/synthetic/` between generation and the first run. I do not know why. Nothing was scored on them. I regenerated both slices with the same commands, spot-checked the new files as above, and scored only the regenerated cases. A copy of the 50 files was kept outside the repository during the runs.

Things to know about the new cases

- **S-143** is both an issuer variant and a joint bill (two names). The label records both names. Added to the hard-cases table.
- **S-140** (05 Nov 2026) and **S-143** (01 Dec 2026) have statement dates after the application date (2026-10-05). The label is still what is printed. This is also why the rules disagree with the placeholder decision on most synthetic cases (see the report).
- **Placeholder decision.** Every synthetic case carries `tom_decision: approve`. Nobody at the bank decided these cases. Rules-versus-Tom is therefore quoted on the 50 real cases only.
- Generated towns and postcodes do not always match (a town like Brightwell with a GL12 postcode). They are fictional, and the label follows the page.

3 · Report to Deborah

Date: 2026-11-06 (runs saved 02:50 UTC) · **Set:** 50 real (Tom's labels) + 50 synthetic · **Prompts compared:** extract, extract_v2 · **Model:** the real model through OpenRouter (`FDE_FAKE_LLM=0`), one run per prompt and case.

Every number below comes from `python -m scripts.run_eval`, saved in `engagement/eval/runs/`.

The headline

With the fixed prompt UC1 reads 98.5% of fields right on 100 cases (97.0% on the 50 real documents, 100% on the 50 synthetic ones), but on real documents the rules still disagree with Tom on 8 of 47 decisions.

The biggest remaining failure is the issuer: the model copies the service tagline into the name ("Harrowgate Water & Sewerage" for "Harrowgate Water") in 4 real cases, and the rules also reject legal-name suffixes and "number, Flat n" addresses that Tom accepts.

Scores

	extract (Day 3)	extract_v2 (Day 4)
All fields, 100 cases	97.8% (391 of 400)	98.5% (394 of 400)
Real cases only (50)	96.0% (192 of 200)	97.0% (194 of 200)
Synthetic only (50)	99.5% (199 of 200)	100.0% (200 of 200)
Rules agree with Tom, real cases	37 of 47	39 of 47

Rules versus Tom is quoted on real cases only. The "all" runs show 46 of 97 and 48 of 97, but the 50 synthetic cases carry a placeholder `approve` that Tom never gave, and 41 of them fail a rule (statement dates months before the application date, legal-name suffixes), so including them would measure the placeholder.

By field (extract_v2, all 100): full name 100%, address 100%, statement date 98%, issuer 96%.

By slice

Both runs, all 100 cases (real and synthetic together, as the slices overlap).

Slice	Cases	extract	extract_v2	Note
two_dates	28	98.2%	99.1%	No date errors in either run; the one miss left is an issuer tagline (D04-043).
joint_names	19	98.7%	97.4%	Both names were read in every case; the two misses are issuer taglines (D04-040, D04-048).
issuer_variant	10	100%	100%	Legal suffixes read correctly; the rules reject them (D04-036, D04-049).
address_order	9	97.2%	100%	Addresses read as printed; the extract miss was a synthetic issuer (S-147 "Broadband Broadband"). The rules reject two real ones (D04-041, D04-045).
clean	23	97.8%	98.9%	Remaining miss is the issuer tagline (D03-023).
name_differs	1	75.0%	100%	Single case; the difference is run-to-run noise (issuer tagline on D03-005).
smudge	3	91.7%	91.7%	One OCR misread: D03-011 read 10 Sep, printed 11 Sep.
ocr_misread	1	75.0%	75.0%	D03-009 read 16 Sep, printed 18 Sep.
date_rule, initials, moved, model_guess, old_document, unapproved_issuer	1 each	100%	100%	One case each; tells you nothing about the slice.

The next fix

The issuer, a prompt first, then a rule. The tagline error hit 7 fields with extract (6 real, 1 synthetic: "Broadband Broadband") and 4 fields with extract_v2 (all real). Fix 1: `extract_v3` with an instruction to give the organisation name

only and stop before the service line (`Water & Sewerage`, `Electricity & Gas`); it lives in `backend/app/prototype/prompts/`. Re-run on the 5 to 7 affected cases plus the 100. Fix 2, in the rules (`engagement/poa_rules.json` and the issuer and address checks in `backend/app/prototype/rules.py`): normalise legal suffixes and compare the first line of an address as a set of numbers, because 4 of the 8 real rule disagreements are Tom-approved legal-name and flat-order cases (D04-036, D04-049, D04-041, D04-045). The other 4 are tagline-polluted issuers and go away with fix 1.

What this evaluation cannot tell you

- **Half the cases are synthetic and text only.** They test the prompt on clean generated words, not the OCR on scans; the model that wrote them is the same family as the one that reads them, so 100% on synthetic is a ceiling, not a promise. Nobody at the bank checked their labels; I did, against the v2 guidelines.
- **50 real cases is small, and each case ran once.** A 3-field difference between the prompts (394 against 391) is inside run-to-run noise: the issuer tagline error landed on different cases in the two runs although the date instruction cannot affect the issuer. The payment-date mistake did not appear on real documents with either prompt, so these runs do not show that extract_v2 fixed it.
- **Two OCR misreads (D03-009, D03-011)** are wrong in both runs; only a second reader or a person can catch them.
- **Rules versus Tom is measured on 47 real decisions** because three cases are routed to a person; for the synthetic cases there is no Tom decision at all.